

FJFI - Department of Physics
02MECH - Mechanics
Test 2

18.12.2024
12:00pm - 1:30pm

Name					
Question	1	2	3	4	Total
Score					

Instructions

1. The time allocated for the exam is 90 minutes.
2. The maximum score for the test is 4 points and each question is worth 1 point.
3. During the exam, you are not allowed to use any class materials or ask your neighbor for help.
4. Calculators and use of any class materials are not allowed.
5. Pencils and erasable pens are not allowed.
6. Write your answers legibly. Illegible answers will not be scored.
7. **The grading of your exams will be based on the answers you have written on the five stapled sheets of paper you are holding in your hand.**

Question 1

Consider a system of three point particles, each having an identical mass m , with their positions defined by time-dependent vectors,

$$\vec{r}_1 = (at^2, vt, 0), \vec{r}_2 = (vt, at^2, b), \vec{r}_3 = (0, b, at^2),$$

where a , b , and v are positive constants.

- a. Find the total external force acting on this system. — *0.4 points*
- b. Find the total torque acting on this system concerning the origin of the coordinate system. — *0.4 points*
- c. Based on the results from parts (a) and (b), determine if total linear momentum and total angular momentum are conserved. — *0.2 points*

Solution:

Question 2

The equation of motion of a damped oscillator is given as $x(t) = e^{-\beta t} x_0 \cos(\omega_1 t)$. How long does it take for the damped oscillator to halve the amplitude of the oscillation? (Givens: $\ln 2 \approx 0.7$, $m = 1\text{ kg}$, $b = 1.4\text{ kg/s}$) — *1 point*

Solution:

Question 3

For a given central force $f(r)$ the orbit of a particle of mass μ can be calculated by integrating the following differential equation,

$$\frac{d^2}{d\theta^2} \left(\frac{1}{r} \right) + \frac{1}{r} = -\frac{\mu}{l^2} r^2 f(r)$$

where l is the angular momentum of the particle. The particle follows an orbit given by $1 + \epsilon \cos \theta = \alpha/r$, where $\alpha \equiv \frac{l^2}{\mu k}$ and $\epsilon \equiv \sqrt{1 + \frac{2El^2}{\mu k^2}}$ are constants.

- a. Find the force law $f(r)$ for the central force in terms of k and r . — *0.8 points*
- b. Find the potential $U(r)$ associated with $f(r)$. — *0.2 points*

Solution:

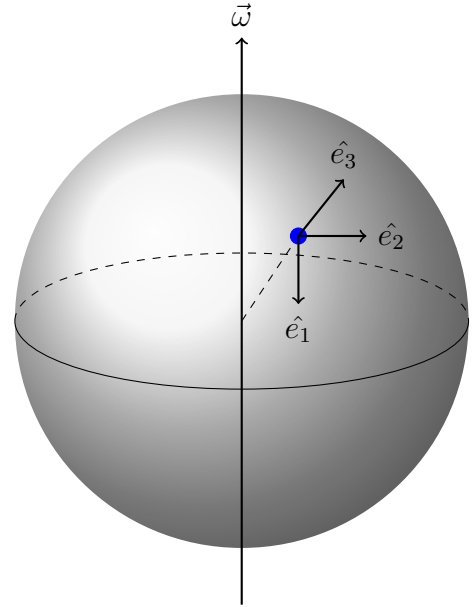
Question 4

In the figure, the earth's rotation about its axis with angular velocity $\vec{\omega}$ is shown. The unit vectors $\hat{e}_1$, $\hat{e}_2$, and $\hat{e}_3$ are local unit vectors on the Earth's surface, where $\hat{e}_3$ is perpendicular to the surface, while $\hat{e}_1$ and $\hat{e}_2$ lie on the surface. The component of $\vec{\omega}$ along the $\hat{e}_3$ direction is ω_z . A particle of mass m is projected horizontally with an initial velocity $\vec{v} = v_0\hat{e}_2$ from the point indicated in the figure. Assume the Earth is a perfect sphere with radius R and neglect air resistance.

a. Determine the direction and magnitude of the Coriolis force. — *0.6 points*

b. Determine the equations of motion in $\hat{e}_1$ and $\hat{e}_2$ directions by considering only the Coriolis force and velocity $\vec{v}$, neglecting free-fall which is in $-\hat{e}_3$ direction. — *0.4 points*

(Given: Coriolis force $\vec{F}_c = -2m\vec{\Omega} \times \vec{v}$)



Solution: